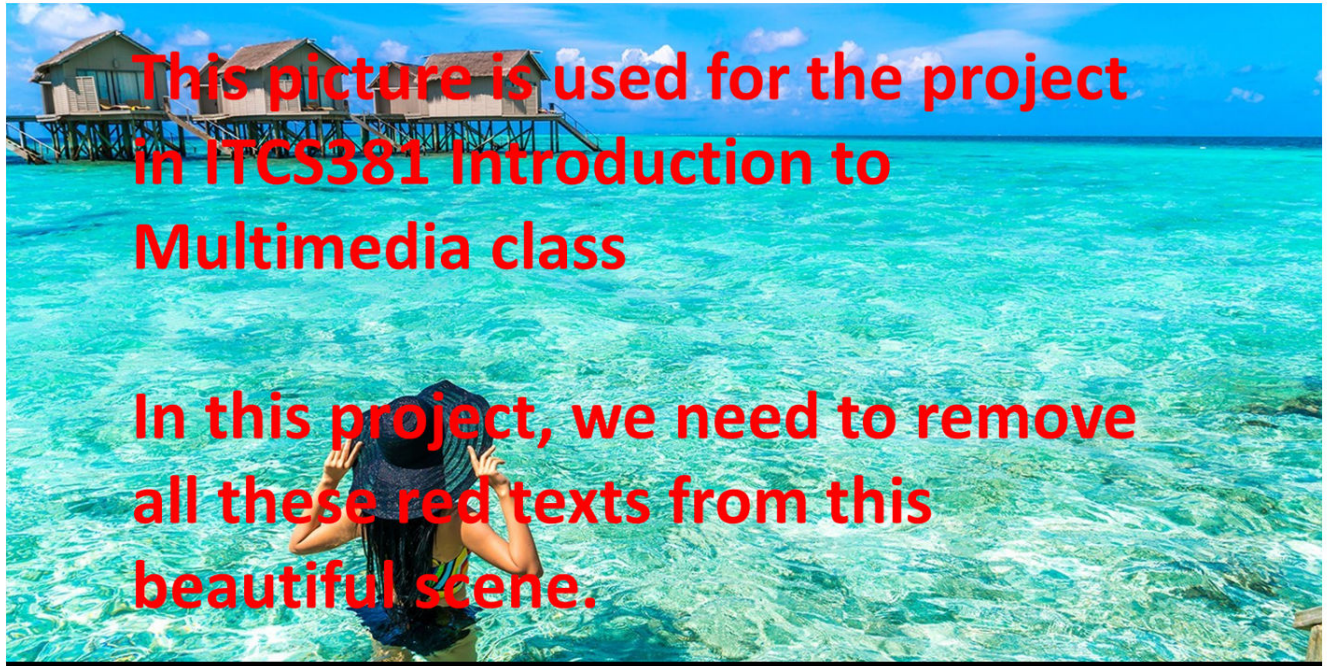


Remove red-text overlay and then remove an object from an image

Read an image "oceanview01_with_text.png" and display it.

```
img = imread("oceanview01_with_text.png");  
[r, c, ch] = size(img);  
imshow(img);
```



Create mask from red-text overlay. In this image, red texts has pure red colors (255, 0, 0). You should also make or expand the mask area to be a bit bigger than the area of red texts. You can use "imdilate" function to expand the mask area a bit.

```
mask = double(img(:, :, 1) == 255 & img(:, :, 2) == 0 & img(:, :, 3) == 0);  
kernel = strel('diamond', 3);  
dilated_mask = imdilate(mask, kernel);
```

Display original mask and expanded mask.

```
montage({mask, dilated_mask});
```

This picture is used for the project
in ITCS381 Introduction to
Multimedia class

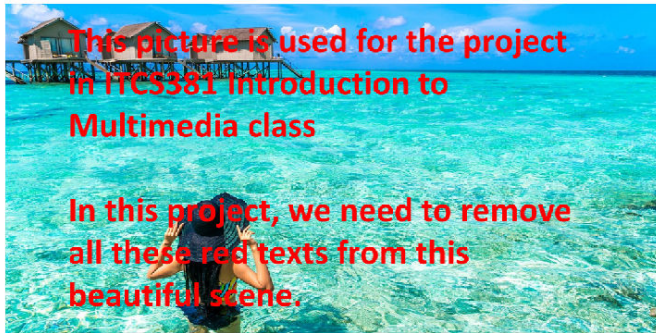
In this project, we need to remove
all these red texts from this
beautiful scene.

**This picture is used for the project
in ITCS381 Introduction to
Multimedia class**

**In this project, we need to remove
all these red texts from this
beautiful scene.**

Display original red-text image and expanded mask.

```
montage({img, dilated_mask});
```



**This picture is used for the project
in ITCS381 Introduction to
Multimedia class**

**In this project, we need to remove
all these red texts from this
beautiful scene.**

Using inpaintCoherent function to remove red-text overlay.

```
notext_img = inpaintCoherent(img, logical(dilated_mask));
```

Display the original red-text image and the removed-red-text image version.

```
montage({img, notext_img});
```

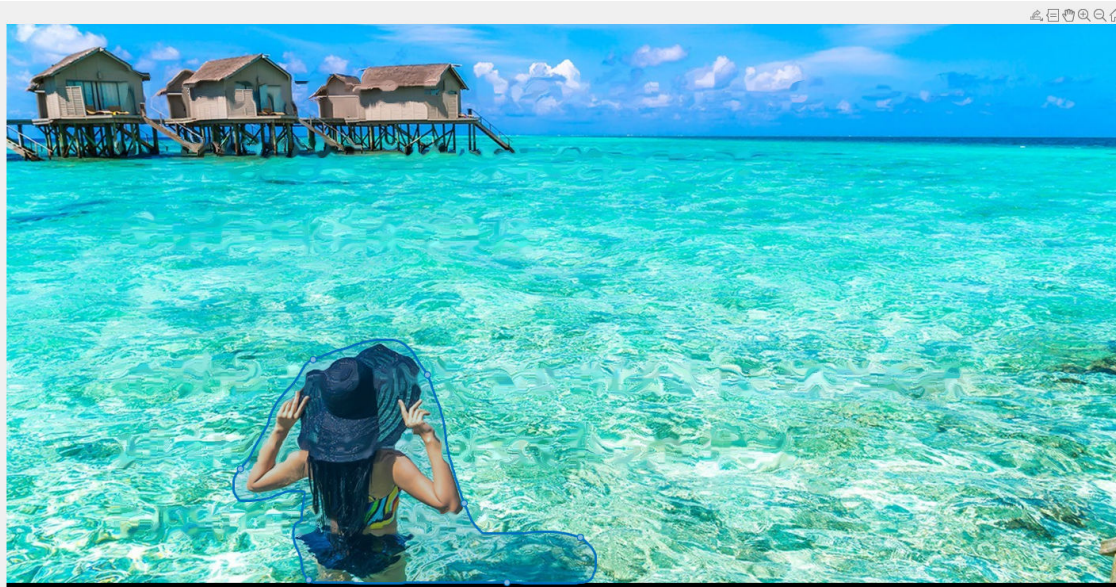



Display only the removed-red-text image version.

```
imshow(notext_img);
```

Use the `drawfreehand` function to select a region of interest (ROI) for inpainting. In this task, you will remove the woman out from the image.

```
h = drawfreehand; % too lazy to write it again
```



Use the `createMask` function to generate a mask from the selected ROIs.

```
freehand_mask = createMask(h);
```

Display the image to be inpainted (the removed-red-text image) and its corresponding mask image.

```
montage({notext_img, freehand_mask});
```

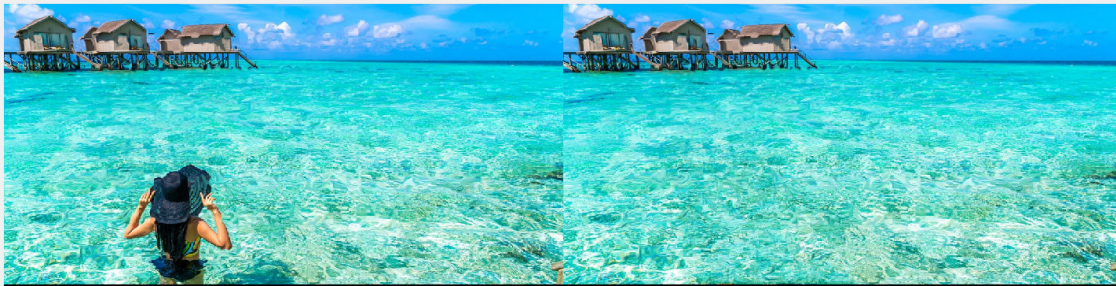


Remove objects in the ROI by using the inpaintExemplar function.

```
nowoman_notext_img = inpaintExemplar(notext_img, freehand_mask);
```

Display the removed-red-text image and the inpainted or the removed-woman image.

```
montage({notext_img, nowoman_notext_img});
```



Display only the final image.

```
imshow(nowoman_notext_img);
```